

Experiment no: 01

Experiment name: Series clipper

Introduction: A series clipper is a fundamental wave-shaping circuit used in electronics to remove or "clip" a portion of an alternative waveform. In a series clipper, the diode is placed in series with the load resistor, meaning the output signal must pass through the diode to reach the load.

Methodology:

- ① When the diode is forward-biased, it acts as a closed switch (ideally), allowing current to flow to output, and vice versa in reverse.
- ② As the diode is in series, the output voltage is only present when the diode is conducting.



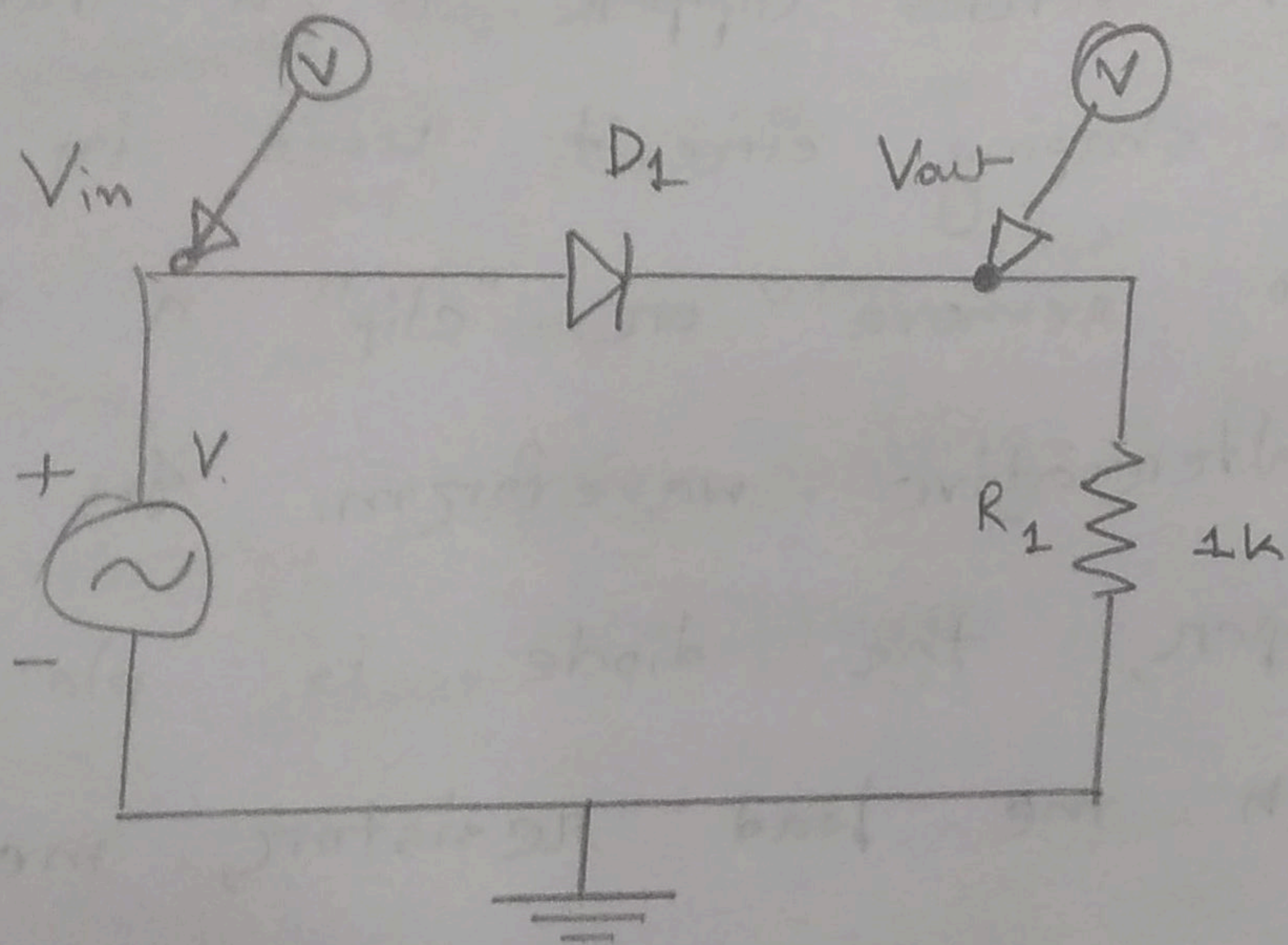


Fig: Series positive clipper

→ Time



Output:

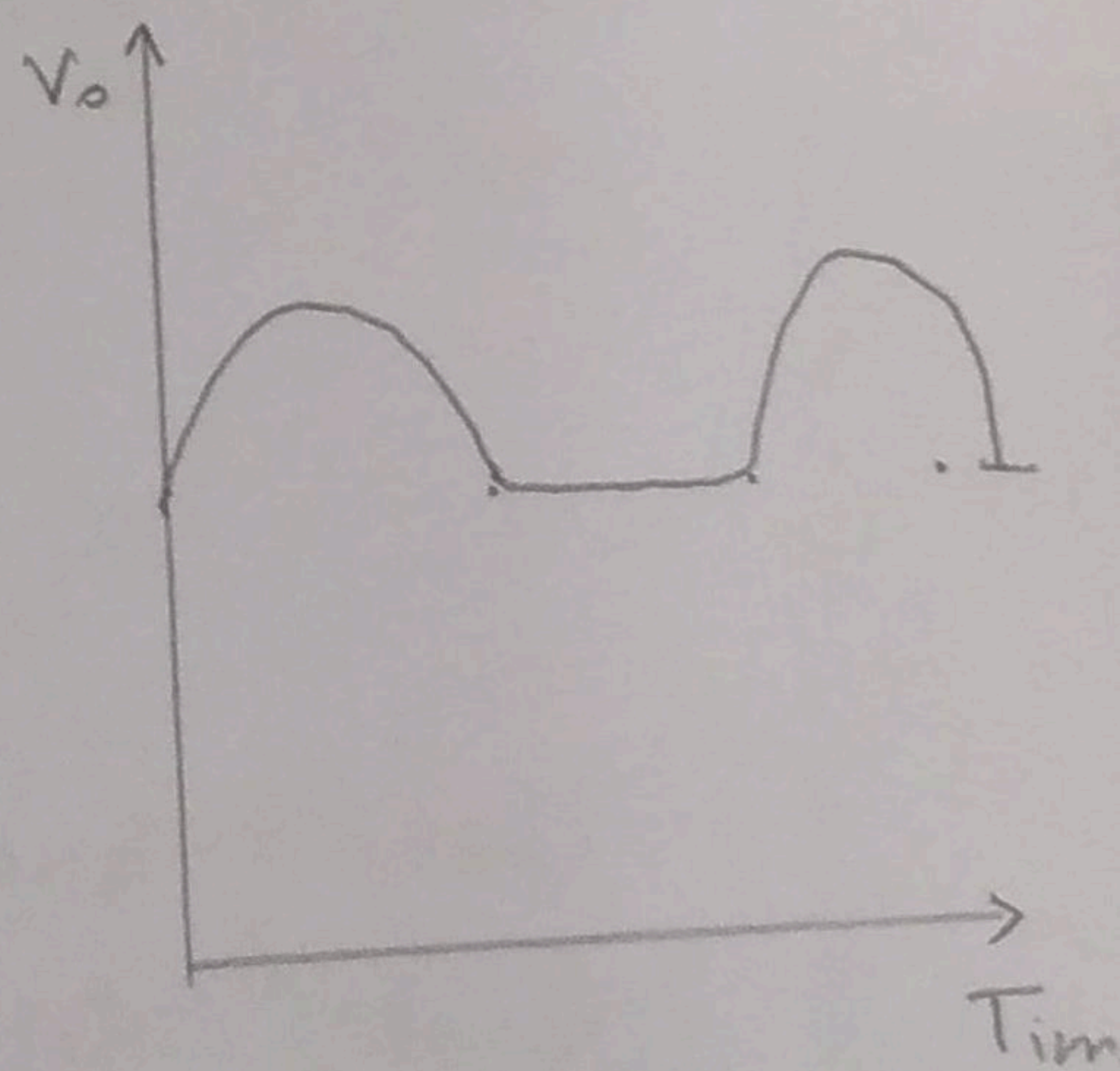
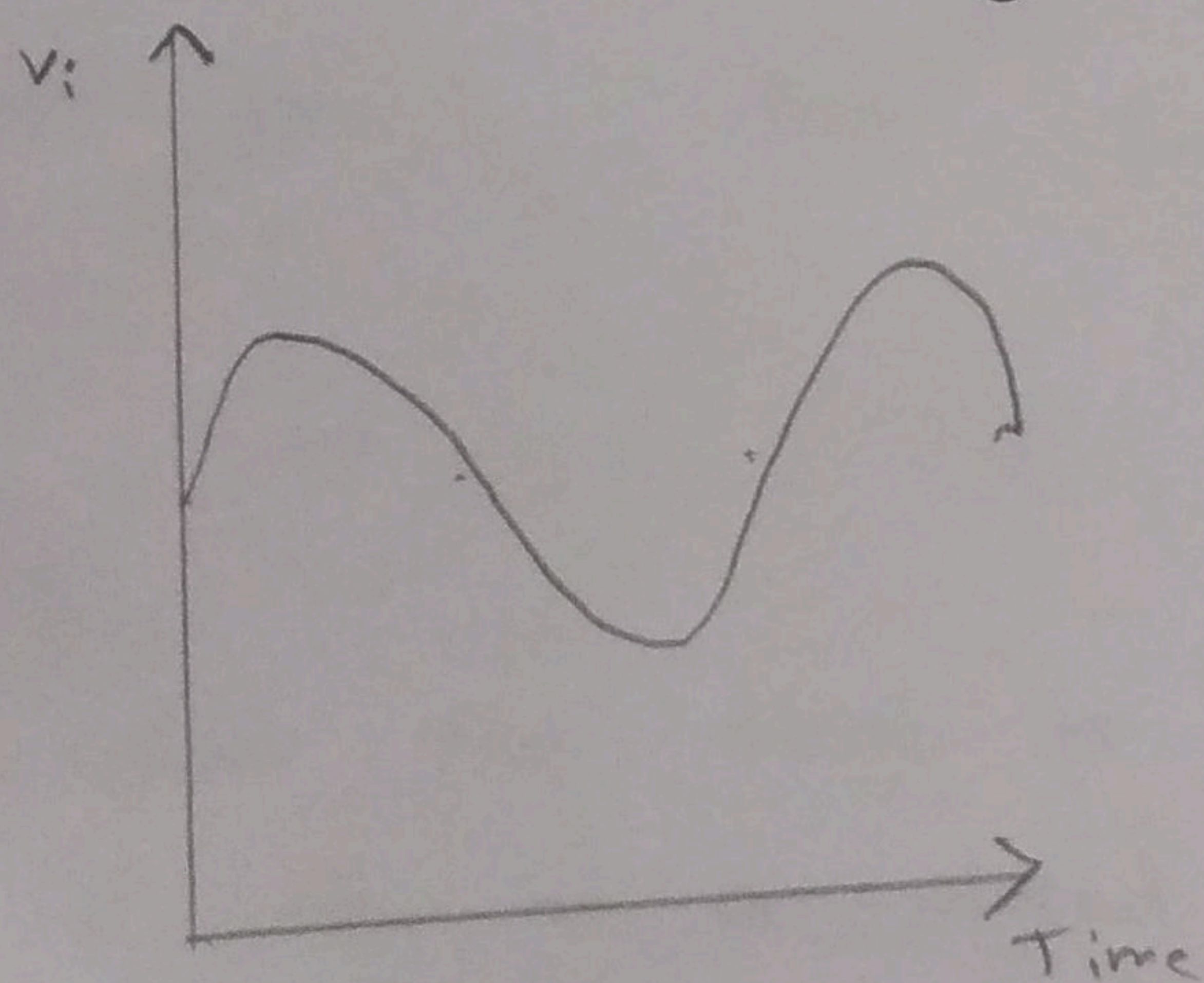
Experimental Procedure: Let's follow the figure and

use PSPICE to draw the circuit,

Later on we use two probes, one for capturing input voltage and the other for output voltage.

Results: After running the simulation, we get

the following output,



Conclusion: Thus we can conclude our simulation model is working as expected.